

UROS-26 Investigating Comparative Performance of Numerical Integrators on a Hamiltonian Two-Body System Under Newtonian Mechanics

Mykyta Khomiakov

June 10, 2026

1 ABSTRACT

The celestial dynamics of the N-body problem are most commonly studied through a restricted set of numerical integration methods, often tracking only a limited range of physical variables. Existing literature focuses predominantly on symplectic integrators, including Leapfrog, Velocity Verlet, Symplectic Euler, and the fourth-order Runge-Kutta method. This paper presents a comparative evaluation of five numerical integration schemes: Explicit Euler, Symplectic Euler, Velocity Verlet, Leapfrog, and Runge-Kutta 4th Order with respect to energy conservation, linear and angular momentum conservation, and computational efficiency within a Hamiltonian framework. Building on this comparison, we further investigate a modification of the best-performing integrator, targeting improvements in energy, angular momentum, and orbital stability for the two-body problem. The trade-offs between numerical precision and computational cost introduced by this modification are examined in the context of a high-performance implementation. Our findings have broader implications for the simulation of celestial mechanics across a range of cosmological settings, with particular emphasis on two-body classical gravitational dynamics.

2 INTRODUCTION

When studying N -body celestial systems, selecting an appropriate numerical integration scheme is fundamental to obtaining accurate and meaningful results. Leapfrog and fourth-order Runge-Kutta (RK4) are routinely employed as standard integrators across modern computational astrophysics, particularly in non-chaotic regimes.

This paper investigates the two-body gravitational problem within the frameworks of Newtonian mechanics and Hamiltonian formalism. In classical mechanics, the Hamiltonian is a function describing the total energy of a system in terms of its generalised coordinates and conjugate momenta, governing the time evolution of the system through Hamilton's equations. This formalism is particularly well-suited to the study of orbital dynamics, as it naturally exposes conserved quantities such as energy and angular momentum against which numerical schemes can be rigorously evaluated.

We restrict our analysis to classical mechanics as defined by Newton and Kepler, applying the governing equations of the field accordingly. A high-performance simulation engine is developed in C++20 to carry out procedurally generated two-body simulations, with the engine exposed to Python 3.12 via a foreign function interface – a common and practical convention in scientific computing workflows. Simulation data is then analysed in Python to draw conclusions regarding integrator accuracy and computational efficiency.

Having identified the most suit-

able standard integrator, we subsequently construct a higher-order scheme and repeat the analysis under equivalent conditions. Final conclusions are drawn on the basis of energy and angular momentum conservation behaviour, alongside a comparison of computational cost across all tested methods.